

Safety Measures of Portable Power Tools

1. Always wear appropriate safety gear.
2. Read instructions in the user manual.
3. Always carry a tool by a grip.
4. Maintain tools with care.
5. Inspect tools before use.
6. Never remove or alter the safety guards.
7. Work on stable surfaces.
8. Be sure the work area is clear of others
9. Unplug power tools when lubricating or changing.
10. Maintain a safe grip on the tool and workpiece.
11. Keep the floor clear of cords and hoses.
12. Only use sharp blades, bits and cutters.

Video:

- <https://youtu.be/tXDBu5-ZAko>

Feedback : 1. Activity

2. Demonstration.

Kwa-Thema Skills School

Subject : Maintenance

Marks : 10

Grade : Year 4

Time : 10 mins

Date : / /2024

Questuion:1

1.State whether the following sentences are True or False.

Statement	True	False
1. Never carry a tool by a grip.		
2. Keep the floor clear of cords and hoses.		
3. Do not wear appropriate safety gear.		
4. Work on stable surfaces.		
5. Remove or alter the safety guards.		
6. Maintain a safe grip on the tool and workpiece.		
7. Inspect tools before use.		
8. Plug power tools when lubricating or changing.		
9. Only use sharp blades, bits and cutters.		
10. Read instructions in the user manual.		

Replace a New Bought Safety Gate.

Video :

- https://youtu.be/AN_hj7G64wM

Steps To Install a New Bought Safety Gate

Step 1 : Measure the gate opening.

Step 2 : Attach right and left frame.

Step 3 : Twist in frame screw.

Step 4 : Place the door in gate frame.

Step 5 : Insert door mounting.

Step 6 : Add extension if needed.

Step 7 : Place spindle rods in each corner.

Step 8 : Position the gate.

Step 9 : Trace each spindle rod.

Step 10 : Mark the center each wall up.

Step 11 : Drill hole for each wall cup.

Step 12 : Screw wall cups into the wall.

Step 13 : Tighten spindle doors.

Step 14 : Tighten until the gap closes.

Step 15 : Rotate the base lock to allow door to swing in one direction.

Feedback : Demonstration Activity

Identification And Demonstration how to use the Hand Held Bending Machine

Steps 1 : Insert the pipe into appropriate bending Steps 2 : Press the handle to apply force even.
groove.



Steps 3 : Bend at any angle required.



Step 4 : Complete the bending.



Video :

- <https://youtu.be/Eo95bjm7WFM>

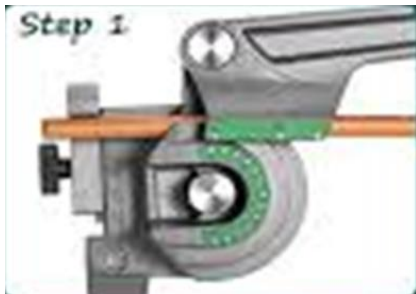
Feedback : 1. Activity

2. Demonstration

Question:1

1. Mention four stages when bending a tube using a handheld Bending Machine.

Stage 1 : _____



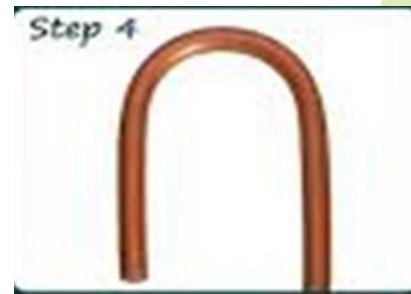
Stage 2 : _____



Stage 3 : _____



Stage 4 : _____



Types of Metal Benders

1. Air Bending



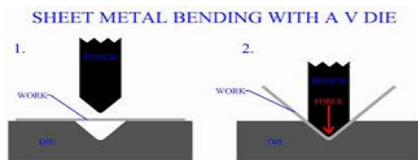
- In this technique, the workpiece is not forced completely onto the die. It provides flexibility and adaptability in terms of bend angles produced with the same tool setup.

3. Coining



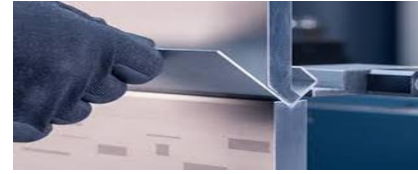
- This is a precision technique where a large amount of force is applied to the workpiece, imprinting the exact angle of the punch and die.

5. V Bending



- It forms a "V" shape and is popular for flexibility.

2. Bottom Bending



- This technique creates a bend by pressing the workpiece fully into the die. The final bend angle and radius mirror the tool shape.

4. Folding



- This technique uses a Folding beam to bend along a straight line axis, commonly used in box and pan bending.

6. Rolling



- This process uses three symmetrically positioned rolls to shape metals.

Feedback : Activity and Demonstration

Question:1

1. Mention the following types of Metal Benders.

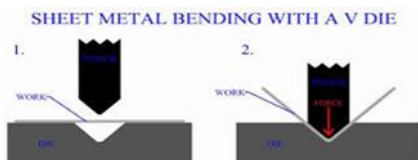
1. _____



3. _____



5. _____



2. _____



4. _____



6. _____



Knowledge And Safety Of Oxygen And Acetylene Equipment



1. Always wear appropriate protective equipment(PPE).
2. Fasten cylinders securely.
3. Never use oil on welding equipment.
4. Open cylinder valves correctly.
5. Keep tip pointed away from your body.
6. Light the flame with the approved lighter.
7. Do not smoke or allow anyone else to smoke near the oxy-fuel gas welder.
8. Do not leave the work area until the cylinder valves are closed.
9. Store oxygen cylinders away from acetylene cylinders.
10. Check connections for leaking gases.
11. Remove regulators and replace protective caps before transporting cylinders.
12. Hand hot metal with a pliers or tongs.

Demonstration Of Basic Brazing Using Gas Equipment

Video:

- https://youtu.be/O_BEf6myVN4

Feedback : Activity

Kwa Thema Skills School

Subject : Maintenance

Marks : 10

Grade : Year 4

Time : 30 mins

Date : / /

Question:1

1. Fill in the missing words in the following sentences.

1.1 Fasten _____ securely.

1.2 Never use oil on _____ equipment.

1.3 Open cylinder _____ correctly.

1.4 Light the flame with the approved _____ .

1.5 Do not leave the work area until the cylinder valves are _____.

1.6 Store oxygen cylinders away from _____ cylinders.

1.7 Check connections for leaking _____.

1.8 Always wear appropriate _____ equipment.

1.9 Hand hot metal with a pliers or _____.

1.10 Remove regulators and replace protective _____ before transporting cylinders.

[welding: lighter: acetylene: protective: caps : cylinders : valves : closed : gases : tongs]

(10)